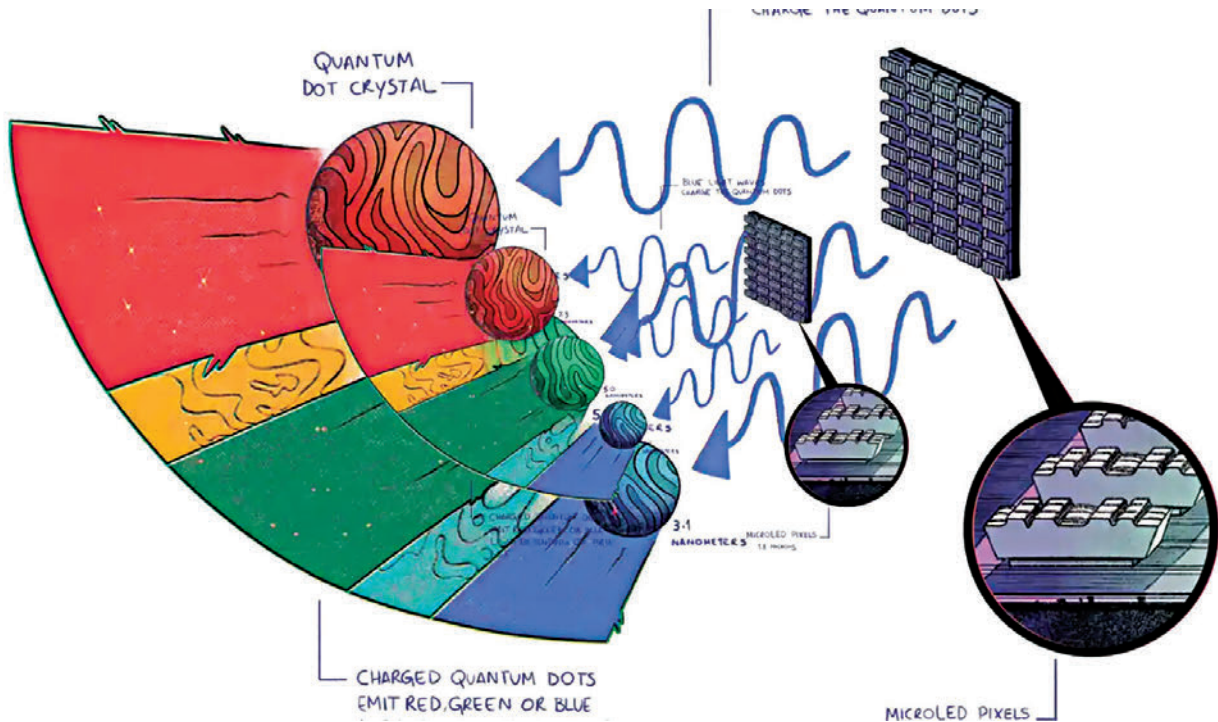




QLED vs QDEL vs QD-OLED vs QD-micro LED

Everything you need to know about the different panel types

Deepak Singh | deepak@digit.in

B quantum Dots have undeniably become the cornerstone of modern TV innovation. These high-efficiency colour converters are being used to enhance all sorts of display technologies including LCD, OLED and even micro LED. Despite their growing presence, there's often confusion about the different types of Quantum Dot-based technologies and how they

differ. Let's break down the most popular types and highlight their differences.

What are quantum dots?

Quantum dots are tiny semiconductor particles that emit light when excited. What colour of light they emit depends on the size of the quantum dot. The important thing is that there is no loss of efficiency (in a way, brightness) when colour conversion takes place via

quantum dots. In comparison, when colour filters are used a significant portion of light is absorbed resulting in loss of quantum efficiency. Quantum dots can be classified into several types based on the material used and applications. But there are two popular types that you need to be familiar with.

Electroluminescent Quantum Dots (EL-QDs) emit light when excited by an electric current. The more popular ones